

Chapter 9 — Occupational Health in Urology — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Musculoskeletal disorders (MSDs):** 46–90% of urologists
- **Burnout:** urology ~50% vs 28% general workforce
- **Elbow 90–120°**, shoulders abducted <30° — every modality
- **Monitor top at/below eye level** → gaze-down ~20°
- **Radiation:** ALARA (as low as reasonably achievable) — time ↓, distance ↑, shielding, collimate
- **Pregnant surgeon:** fetal dosimeter at waist, under apron
- **CO₂ laser condylomata** → N95 + evacuator tip ≤2 in (5 cm)
- **Intravesical chemo** → double chemo gloves, gown, shield, goggles
- **Exercise:** 150 min moderate or 75 vigorous + 2 days strength/week
- **Sleep:** 7–9 h (25–64 y); 7–8 h (≥65 y)

PHYSICAL ERGONOMICS

- **Open:** working area 2–4 in below elbow; neck flexion <25°; set for tallest, steps for others
- **Laparoscopic:** table at surgeon's pubic symphysis → handles at/below elbow
- **Endourology:** target between surgeon's symphysis and umbilicus
- **Robotic console:** back flexion <15°; knee 90° rest, max 120° at pedals; forearm parallel to floor
- **Loupes:** steep declination, neck <20°; lightest available
- **Lead:** two-piece or belted one-piece → weight on hips, not shoulders
- **Lifting:** ≥2 people; pivot, never twist
- **Microbreaks/stretching:** ↓ pain, ↑ focus, no added case time
- **Workstation:** knees 90°, elbows 90°, screen 20–40 in (50–100 cm)
- **Presbyopia:** ~10 years into practice; "workspace" (near + intermediate) lenses; multifocals → lower monitor
- **Non-modifiable MSD risks:** young, inexperienced, short, small glove, female, high body mass index

OCCUPATIONAL EXPOSURES

- **Vaccinate staff (CDC):** hepatitis B, influenza, measles-mumps-rubella
- **Bloodborne:** human immunodeficiency virus, hepatitis B and C via sharps → safety devices, reporting, post-exposure protocol
- **Smoke:** particulates, toxic gases, bioaerosols → evacuation first
- **Chemo spills:** closed-system drug-transfer device
- **Radiation harms:** cataract, infertility, cancer; dosimeters mandatory
- **Fluoroscopy:** low-dose default; pulsed, lowest rate (1 pulse/s); last-image hold
- **Geometry:** source far, image intensifier close to patient
- **Surgeon foot-pedal control:** spot images, not continuous
- **Pregnancy:** maternity/double lead; minimise radiation cases
- **Noise:** above concentration limits, usually below hearing-damage levels
- **Protect critical steps:** music off, no non-urgent pages, no breaks

PHYSICAL HEALTH OF SURGEONS

- **Deaths 25–54 y:** heart disease, unintentional injury, suicide all top five
- **Most threatening modifiable cause 35–64 y:** ASCVD (atherosclerotic cardiovascular disease)
- **Surgeons use primary care least** — time, self-sacrifice, self-treatment, fear of judgement
- **Intraoperative stress response:** heart rate ↑, heart-rate variability ↓, cortisol ↑ → baseline ~30 min
- **Sleep deprivation:** performance ↓ ≥12%; ↑ fatal car crash

- **Biomarkers:** low-density lipoprotein, DEXA (dual-energy X-ray absorptiometry) body composition, VO₂ max, oral glucose tolerance test

MENTAL AND EMOTIONAL HEALTH OF SURGEONS

- **Burnout triad** (Freudenberger 1980): exhaustion, depersonalisation, low accomplishment
- **Burnout, all medical providers:** 40–60%
- **Urology burnout risk:** more hours/encounters; **mid-career > early;** chronic MSD pain
- **≥20% time on meaningful work** → less burnout
- **Second victim** → "psychological first aid", peer support
- **Stress continuum:** Ready (green) → Reacting (yellow) → Injured (orange) → Ill (red)
- **Ill zone** → professional help, sick leave
- **Cognitive behavioural therapy (CBT) and mindfulness:** ↑ resilience (trials); dialectical behavioural therapy = CBT + acceptance
- **Imposter syndrome:** cognitive distortion, commonest in trainees → CBT
- **Distortion → tactic:** overgeneralisation/depersonalisation → self-compassion; catastrophising/minimisation → double standard; labelling/tunnel vision → exploratory questions; emotional reasoning → restructuring; confirmation bias → defusion

CLASSIC TRAPS

- **Trap:** laparoscopic table set to elbow — **Actually:** symphysis; handles at elbow
- **Trap:** one-piece lead is the problem — **Actually:** shoulder loading; belt it
- **Trap:** magnification harmless — **Actually:** increases dose
- **Trap:** intensifier far from patient — **Actually:** close; source far
- **Trap:** microbreaks lengthen cases — **Actually:** no added time
- **Trap:** burnout peaks early career — **Actually:** mid-career higher
- **Trap:** burnout = "Ill" zone — **Actually:** Injured (orange)
- **Trap:** self-care fixes burnout — **Actually:** system change essential
- **Trap:** progressives fine for desk — **Actually:** lower monitor or workspace lenses
- **Trap:** "second victim" universally accepted — **Actually:** may offend; "traumatised caregiver"